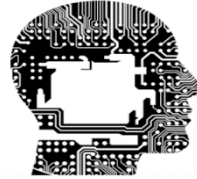




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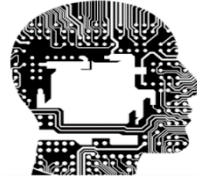
Apprentissage par Machine 02

APM2

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Apprentissage par Machine APM 02

Workshop 04

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Etudiants concernés

Faculté/Institut	Département	Niveau	Spécialité
New Technologies	IFA	Master 01	STIC

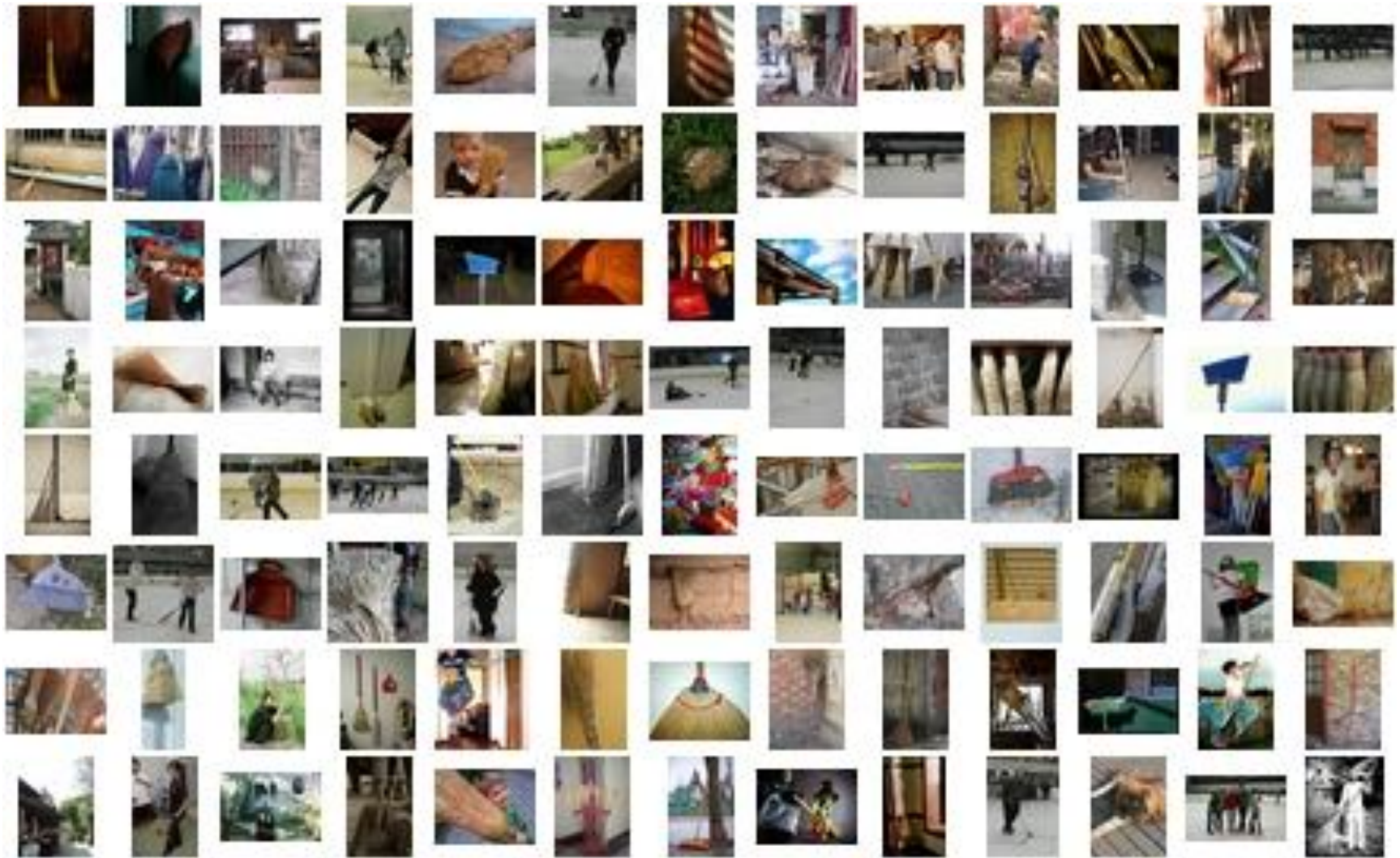
- CNN application: Image Identification

- Task: identify random images from the ImageNet dataset

- About ImageNet

- A dataset containing over 10 million labeled images
- Created by the ImageNet organization for computer vision research
- Example samples are shown in the next slide

Convolutional Neural Network



Convolutional Neural Network

- CNN application: Using the VGG CNN model
- Task: classify three images using the VGG (Visual Geometry Group) CNN model



- Before starting
 - Install the necessary packages using the `pip` command

Convolutional Neural Network

- CNN application: step-by-step guide
 - Step 0: import all the required libraries

```
C: > Users > KHALED > Downloads > Perceptron-master > Perceptron-master >  cnn.py > ...  
1  from keras.preprocessing.image import load_img  
2  from keras.preprocessing.image import img_to_array  
3  from keras.applications.vgg16 import preprocess_input  
4  from keras.applications.vgg16 import decode_predictions  
5  from keras.applications.vgg16 import VGG16  
6
```

Convolutional Neural Network

- CNN application: step-by-step guide
 - Step 1: Load the pre-trained VGG16 model

```
C: > Users > KHALED > Downloads > Perceptron-master > Perceptron-master >  cnn.py > ...
```

```
6
```

```
7     model = VGG16()
```

```
8
```

Convolutional Neural Network

- CNN application: step-by-step guide
 - Step 2: Import the three images you want to classify

```
C: > Users > KHALED > Downloads > Perceptron-master > Perceptron-master > cnn.py > preprocess
```

```
10
```

```
11  img1 = load_img('C:/Users/KHALED/OneDrive/Bureau/reaper.jpeg', target_size=(224, 224))
```

```
12  img2 = load_img('C:/Users/KHALED/OneDrive/Bureau/loup.jpeg', target_size=(224, 224))
```

```
13  img3 = load_img('C:/Users/KHALED/OneDrive/Bureau/ballon.jpeg', target_size=(224, 224))
```

```
14
```


Convolutional Neural Network

- CNN application: step-by-step guide
 - Step 3: preprocess the images to match the input requirements of VGG16

```
C: > Users > KHALED > Downloads > Perceptron-master > Perceptron-master > cnn.py > pred_modele
14
15 # Conversion image (matrice de pixels) en un numpy array
16 def preprocess(image) :
17     image = img_to_array(image)
18
19     # Redimensionnage
20     image = image.reshape((1, image.shape[0], image.shape[1], image.shape[2]))
21
22     # Preprocessing
23     image = preprocess_input(image)
24
25     return image
```

Convolutional Neural Network

- CNN application: step-by-step guide
 - Step 4: run the model's prediction on the preprocessed images

C:\> Users > KHALED > Downloads > Perceptron-master > Perceptron-master > cnn.py > ...

```
27 def pred_modele(image) :
28
29     image = preprocess(image)
30     # Prédiction
31     y_pred = model.predict(image)
32     # Conversion des probabilités en classe label
33     label = decode_predictions(y_pred)
34     # Affectation du label ayant la plus grande probabilité
35     label = label[0][0]
36     return ((label[1], label[2]*100))
37 img=[img1,img2,img3]
38
39 for i in range(3) :
40     print("Prédiction image",i+1,":",pred_modele (img[i])[0], 'avec une probabilité de',round(pred_modele (img[i])[1],2),'%')
41
```

Convolutional Neural Network

- CNN application: step-by-step guide
 - Step 5: display the prediction results

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
2024-03-05 12:33:07.846776: W tensorflow/tsl/framework/cpu_allocator_impl.cc:83] Allocation of 411041792 exceeds 10% of free system memory.
1/1 [=====] - 1s 843ms/step
1/1 [=====] - 0s 342ms/step
Prédiction image 1 : warplane avec une probabilité de 45.5 %
1/1 [=====] - 0s 333ms/step
1/1 [=====] - 0s 344ms/step
Prédiction image 2 : timber_wolf avec une probabilité de 94.48 %
1/1 [=====] - 0s 360ms/step
1/1 [=====] - 0s 330ms/step
Prédiction image 3 : soccer_ball avec une probabilité de 99.99 %
1/1 [=====] - 0s 348ms/step
PS C:\Users\KHALED> |
```